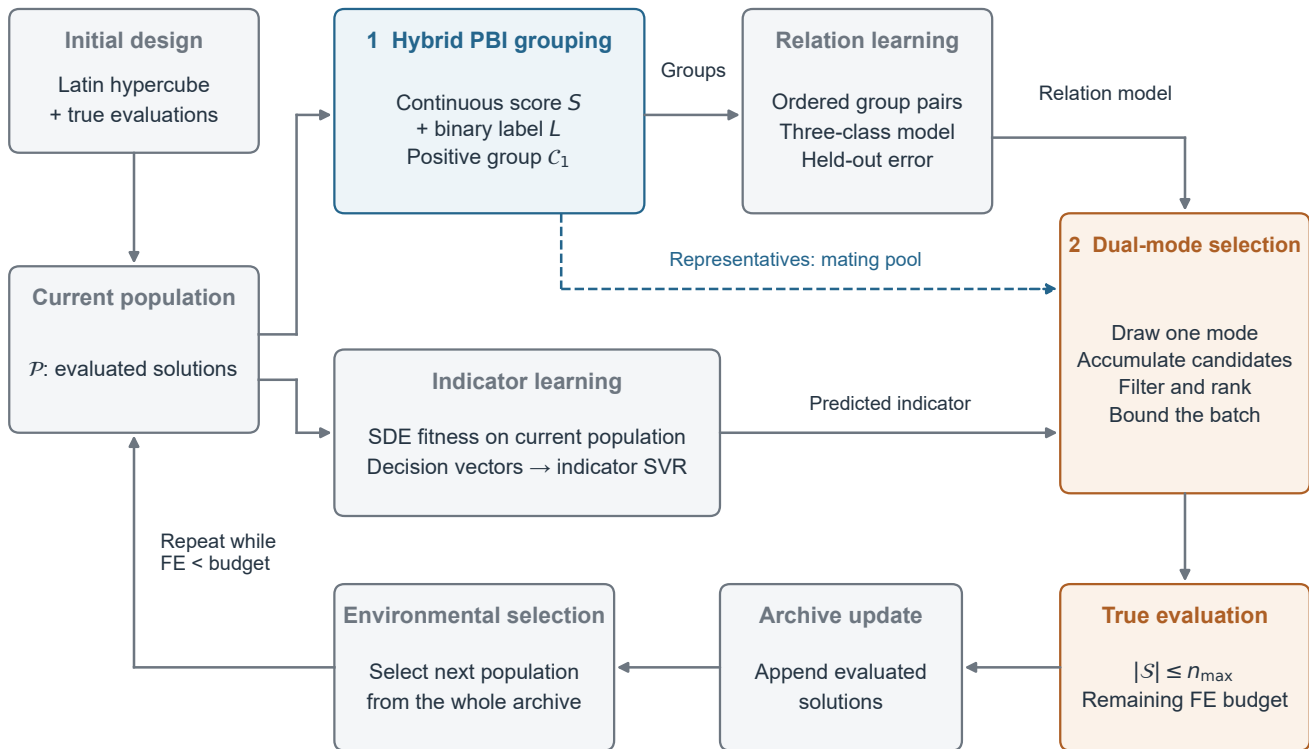
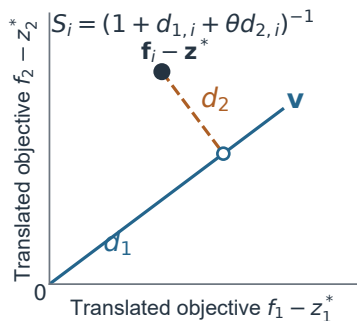
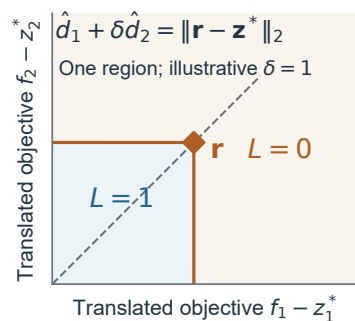
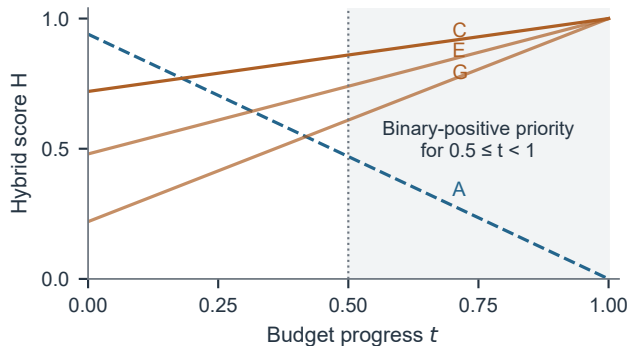




a Continuous directional quality**b Representative-relative boundary****c Hybrid score: $H = (1-t)S + tL$** **d Top-quarter group from eight solutions**

t=0	0.94	0.83	0.72	0.61	0.48	0.36	0.22	0.10
t=0.25	0.70	0.62	0.79	0.46	0.61	0.27	0.42	0.33
t=0.5	0.47	0.41	0.86	0.30	0.74	0.18	0.61	0.55
t=0.75	0.23	0.21	0.93	0.15	0.87	0.09	0.81	0.78
	A L=0	B L=0	C L=1	D L=0	E L=1	F L=0	G L=1	H L=1

Outlined cells enter the positive group (2/8).

Analytical illustration of the definitions; values and geometry are not experimental observations.

Draw one mode per outer iteration

Indicator available: each mode has probability 0.5; otherwise use exploration.

Exploration: relation-guided search

Sharpness-weighted pair averages $\rightarrow$ variation
Accumulate offspring; deduplicate the candidate pool

Acquisition and quantile filtering

$A_{\text{exp}} = \bar{R} + \lambda_t \bar{U}$
Weight decreases with progress and model error

Greedy batch construction

Acquisition quality + distance from selected decisions

Indicator: relation-guided search

Unweighted pair averages $\rightarrow$ variation
Accumulate offspring; deduplicate the candidate pool

Relation filter $\rightarrow$ indicator reranking

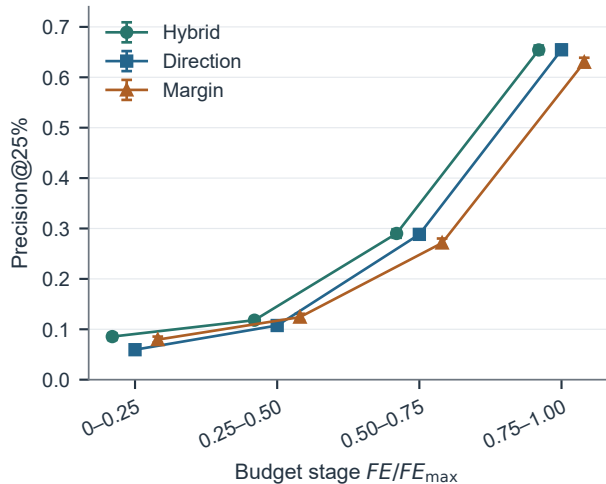
Top relation scores $\rightarrow$ predicted indicator $\hat{I}$
Retain candidates above the indicator quantile

Ordered batch construction

Select by descending predicted indicator

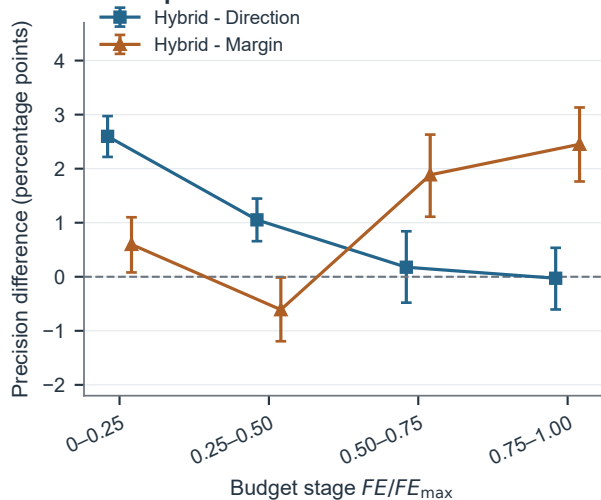
Retained-set and remaining-budget bound: $|S| = \min(n_{\max}, |\mathcal{H}^m|, FE_{\max} - FE)$

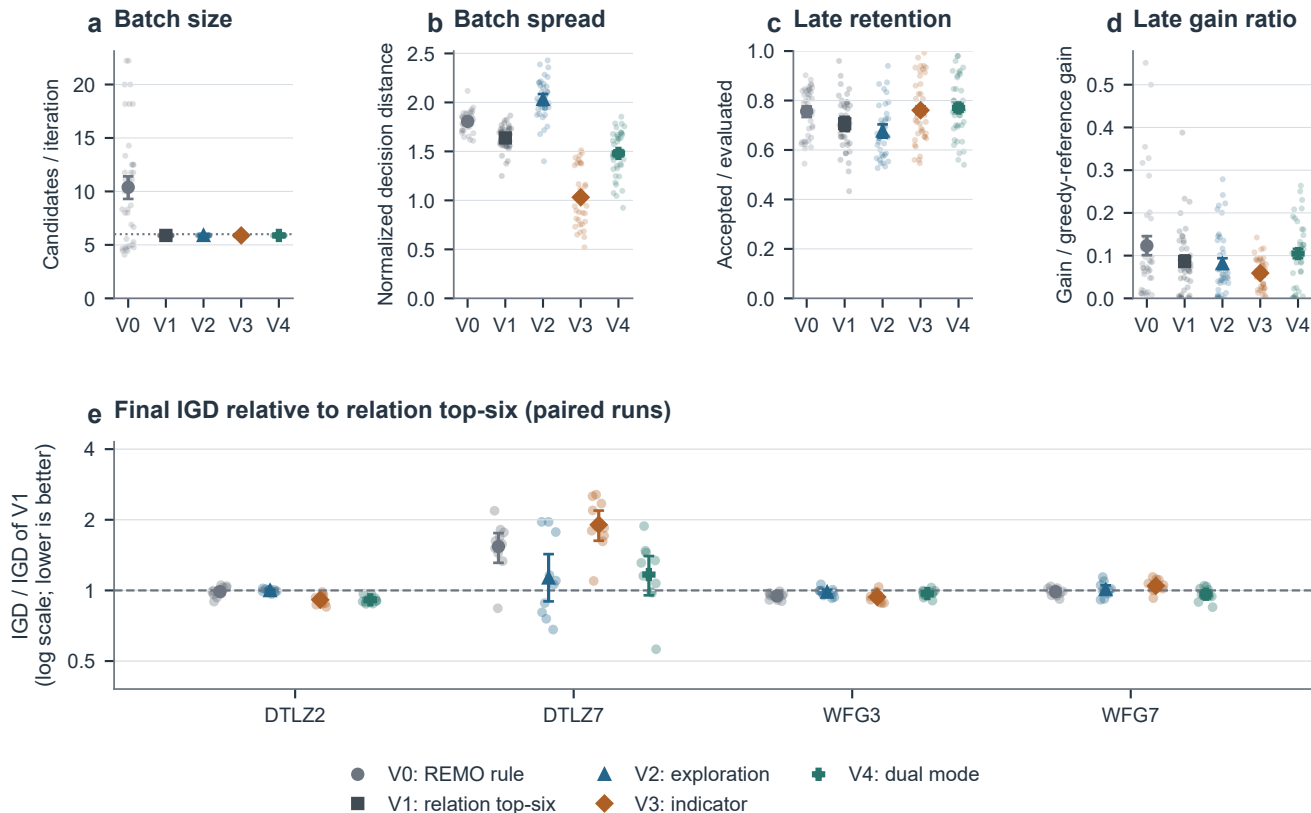
a Final-population retention



Means and 95% stratified bootstrap CIs; checkpoints averaged within each run and stage.

b Paired precision differences





a–d: 40 run values per policy; means and 95% stratified bootstrap CIs.

e: 10 paired IGD ratios per problem and policy; geometric means and 95% bootstrap CIs.

a **M = 10**

DTLZ1	1.11	0.95	0.98	0.84	0.62*	0.62*	1.03
DTLZ2	1.00	1.13	1.00	0.75	0.71	0.71	0.83
DTLZ3	1.16	1.69	1.63	1.67	0.70	0.69	1.04
DTLZ4	1.03	0.95	0.95	0.74	0.68	0.68	0.97
DTLZ5	1.06	2.28	1.76	0.67	0.67	0.66	0.86
DTLZ6	0.88	2.64	1.61	1.10	0.80	0.80	0.90
DTLZ7	0.57	0.65	0.59	1.95	0.42	0.42	0.44
WFG1	0.99	0.96	0.92	1.00	0.93	0.93	0.98
WFG2	0.60	0.89	0.61	0.94	0.59	0.55	0.60
WFG3	1.12	1.44	1.35	1.07	1.03	1.03	1.08
WFG4	0.81	1.15	0.93	1.15	0.73	0.72	0.73
WFG5	0.86	1.04	0.90	0.91	0.84	0.86	0.84
WFG6	0.82	0.89	0.73	1.00	0.72	0.72	0.71
WFG7	0.84	0.77	0.65	0.82	0.68	0.67	0.72
WFG8	0.81	0.90	0.76	1.02	0.76	0.76	0.73
WFG9	0.83	1.10	0.90	0.99	0.82	0.82	0.78

REMO PIEA MCEAD R2AEA PCSAEA KRVEA CSEA

b **M = 20**

DTLZ1	1.18	0.85	1.04	0.73	0.51	1.00	0.48
DTLZ2	1.02	0.98	0.93	0.86	0.84	0.96	0.84
DTLZ3	1.07	1.10	2.35	0.93	0.55	1.12	0.53
DTLZ4	1.02	0.94	0.82	0.86	0.84	1.00	0.84
DTLZ5	1.12	1.99	2.00	0.68	0.73	1.05	0.74
DTLZ6	0.91	3.08	1.62	1.19	0.72	0.90	0.72
DTLZ7	0.47	0.97	0.64	1.29	0.36	0.37	0.37
WFG1	1.00	0.98	0.97	1.00	0.98	0.99	0.98
WFG2	0.68	0.90	0.71	1.18	0.65	0.62	0.62
WFG3	1.21	1.60	1.37	1.05	1.09	1.18	1.11
WFG4	0.82	1.16	1.17	1.33	0.81	0.81	0.81
WFG5	0.89	1.04	1.00	0.97	0.92	0.88	0.92
WFG6	0.79	0.84	0.75	1.22	0.76	0.73	0.75
WFG7	0.85	0.84	0.75	1.00	0.78	0.78	0.79
WFG8	0.83	0.89	0.80	1.27	0.81	0.77	0.80
WFG9	0.88	1.02	1.00	1.02	0.89	0.83	0.88

REMO PIEA MCEAD R2AEA KRVEA CSEA HES-EA



0.5 1 2

Mean IGD of HPDC-MaOEA / comparator ($\log_2$ color scale)

Ratio < 1 favors HPDC-MaOEA. * Identical source entries awaiting provenance check; no significance encoded.